

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2 (APPLICATION BASED PROBLEM SOLVING)**

**COURSE CODE:CSA09**

**COURSE NAME: Programming in  
JAVA**

### **COURSE OUTCOME COVERED**

**CO5.** *Construct database-driven web applications by implementing JDBC connectivity to perform SQL operations such as insert, update, and delete using different statement types. (BL3, BL6)*

Assessment Tool Weightage: 20%

**QUESTIONS: 1**

**Total Marks:50**

Annexure A

### **APPLICATION BASED PROBLEM SOLVING**

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#### **Code of Conduct:**

*I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:**

## Annexure A

### APPLICATION BASED PROBLEM SOLVING

#### 1: Smart Library Book Registration System (Statement + PreparedStatement)

A college library is digitizing its book inventory. The librarian needs a Java application that allows new books to be added while preventing duplicate ISBN numbers from being stored.

Database Table: Books

**BookID Title Author ISBN AvailableCopies**

Tasks:

1. Establish JDBC connectivity with MySQL.
2. Insert five book records using PreparedStatement.
3. Before insertion, verify whether the ISBN already exists using a Statement.
4. Display an appropriate success or duplicate message.

Problem-solving focus: Choosing between Statement and PreparedStatement for validation and insertion.

#### 2: Hospital Patient Record Updation System (PreparedStatement)

A hospital maintains patient records in a database. Doctors frequently update contact numbers and diagnosis information after each consultation.

Database Table: Patients

**PatientID Name Phone Diagnosis LastVisit**

Tasks:

1. Connect the Java application to the database using JDBC.
2. Accept a Patient ID from the user.
3. Update the phone number and diagnosis using PreparedStatement.
4. Display the updated patient record after modification.

Problem-solving focus: Secure parameterized update operations.

| Criteria                            | Weightage | Level 4 (Exemplary)                                                                                                      | Level 3 (Proficient)                                                          | Level 2 (Developing)                                                         | Level 1 (Beginning)                                                        |
|-------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Problem Analysis & Solution Design  | 20%       | Accurately analyzes the application scenario and designs a complete JDBC-based solution with appropriate SQL operations. | Correctly analyzes the scenario with minor gaps in the solution design.       | Partially understands the scenario and proposes an incomplete solution.      | Unable to identify the problem requirements or design a suitable solution. |
| JDBC Connectivity & Statement Usage | 25%       | Correctly establishes database connectivity and effectively uses the appropriate JDBC statement type (Statement,         | Connection and statement implementation are mostly correct with minor errors. | JDBC connectivity is partially implemented; statement usage is inconsistent. | Fails to establish database connectivity or use JDBC statements correctly. |

|                                    |     |                                                                                                                             |                                                                         |                                                                        |                                                                          |
|------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|
|                                    |     | PreparedStatement, or CallableStatement ).                                                                                  |                                                                         |                                                                        |                                                                          |
| SQL Operations & Data Integrity    | 20% | Implements INSERT, UPDATE, and DELETE operations accurately while maintaining data integrity and proper parameter handling. | SQL operations are correct with minor logical or syntax issues.         | Performs basic SQL operations with noticeable errors in data handling. | SQL operations are incorrect or fail to modify the database as required. |
| Program Logic & Exception Handling | 20% | Demonstrates strong logical reasoning, validates inputs, and handles exceptions effectively using appropriate techniques.   | Good program logic with basic input validation and exception handling.  | Limited logical flow with minimal validation or error handling.        | Poor program logic with no meaningful exception handling.                |
| Code Quality & Output Verification | 15% | Produces clean, modular, well-documented code and correctly verifies database results through meaningful output.            | Code is organized with minor documentation or output verification gaps. | Code structure is inconsistent; output verification is incomplete.     | Code is poorly structured with incorrect or missing output verification. |

CODE:

```
import java.sql.*;
```

```
public class LibraryBooks {
```

```
    public static void main(String[] args) {
```

```
        String url = "jdbc:mysql://localhost:3306/LibraryDB";
```

```
        String user = "root";
```

```
        String password = "root";
```

```
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con = DriverManager.getConnection(  
                url, user, password);
```

```
            System.out.println("Database Connected Successfully");
```

```
            Statement stmt = con.createStatement();
```

```
            String sql = "INSERT INTO Books " +
```

```
                "(Title, Author, ISBN, AvailableCopies) " +
```

```
"VALUES (?, ?, ?, ?)";
```

```
PreparedStatement ps =
```

```
    con.prepareStatement(sql);
```

```
String[][] books = {
```

```
    {"Python Essentials", "Guido",
```

```
    "ISBN201", "5"},
```

```
    {"C++ Mastery", "Bjarne",
```

```
    "ISBN202", "4"},
```

```
    {"Algorithms Unlocked", "Thomas",
```

```
    "ISBN203", "6"},
```

```
    {"Operating Systems", "Silberschatz",
```

```
    "ISBN204", "3"},
```

```
    {"Web Development", "Jon",
```

```
    "ISBN205", "7"}  
};
```

```
for (int i = 0; i < books.length; i++) {
```

```
String isbn = books[i][2];
```

```
String check = "SELECT ISBN FROM Books " +  
    "WHERE ISBN = '" + isbn + "'";
```

```
ResultSet rs = stmt.executeQuery(check);
```

```
if (rs.next()) {
```

```
    System.out.println(  
        "Duplicate ISBN: " + isbn);
```

```
} else {
```

```
    ps.setString(1, books[i][0]);  
    ps.setString(2, books[i][1]);  
    ps.setString(3, books[i][2]);  
    ps.setInt(4,  
        Integer.parseInt(books[i][3]));
```

```
    ps.executeUpdate();
```

```
    System.out.println(  
        "Book inserted successfully: "  
        + books[i][0]);
```

```
    }

    rs.close();
}

ps.close();
stmt.close();
con.close();

System.out.println(
    "Database connection closed.");

} catch (Exception e) {

    System.out.println(e);
}
}
}
```

OUTPUT:

```
+-----+-----+-----+-----+-----+
| BookID | Title           | Author      | ISBN    | AvailableCopies |
+-----+-----+-----+-----+-----+
| 1 | Python Essentials | Guido       | ISBN201 | 5 |
| 2 | C++ Mastery       | Bjarne      | ISBN202 | 4 |
| 3 | Algorithms Unlocked | Thomas     | ISBN203 | 6 |
| 4 | Operating Systems | Silberschatz | ISBN204 | 3 |
| 5 | Web Development   | Jon         | ISBN205 | 7 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```